

One Imperfect Page per Hundred: I Ambushed My Pipeline with Data Nobody Tuned For

When I moved my pipeline off its own benchmark, brand-name reading accuracy fell from roughly 80% to **38%**.

The share of pages flagged for human review rose from 28% to **97%**.

And exactly **one imperfect page in a hundred** reached the record without a person looking at it.

This post is about why I count that as a pass.

Where this started

In [part 1](#), I built Scribe — a local pipeline that turns handwritten clinical forms into structured data and, more importantly, learned when *not* to trust a model's answer. The headline was that model confidence carried no signal, and disagreement between two independent reads did.

But every number in part 1 had a quiet weakness.

I curated the gold answers. I wrote the extraction schema. I tuned the prompts against the same 23 pages I was scoring on. Even honestly done, that's a loop — the system and its examiner grew up together.

So I went looking for an exam nobody in the room had written.

The ambush

I found a [public dataset of 100 handwritten Indian prescriptions](#), where the ground truth — the list of medicines on each page — was written by the dataset's author, not by me.

The rules I set for myself:

- the extraction schema gets written once, from the task description, before seeing results
- the prompts stay frozen exactly as they were in part 1
- the author's answer conventions are the answer key, not mine

Two things make this set genuinely hostile. The handwriting is real doctors' cursive — not the careful volunteer writing my benchmark was built from. And the drug names are Indian brand names the models have little prior for.

Which is the point. A system built for clinics in one country will eventually meet a register from another.

The reading collapsed

Brand names came back right 38% of the time.

CEPODEM became **Capelin**. *ESOTAB* became **Esotrib**. *NIVEOLI* became **Ulfah Niveoli**.

These are the same confident-cursive misreads that plagued patient names in part 1 — at triple the rate, because the handwriting is harder and the vocabulary is foreign.

I ran both of my best configurations, which differ only in which model does the second read. They produced **identical reading results**. Same primary model, same eyes; the verifier only changed how much got flagged.

That repeats part 1's central lesson in harsher light: **verification cannot fix reading. It can only catch it.**

No prompt I own fixes the reading either. For the local vision models I tested, hard cursive still looks like a real ceiling — and a [recent benchmark on South African maternity records](#) ran into a similar wall even with frontier cloud models.

But the system knew

Here's the part that mattered to me.

Faced with input it couldn't read, the pipeline didn't pretend. On these harder pages, the models were much less willing to claim certainty: most confidence scores fell to around 0.60–0.70, and **97–99% of pages were sent for human review.**

For this input, “a person must look at this” is the correct output.

Auditing the escapes

Three pages out of a hundred skipped human review. I checked every one by hand.

- Two were **fully correct** — single-medicine prescriptions, read cleanly.
- One was an eleven-medicine page that got ten of them right — and **silently dropped one** (BETNESOL INJ).

One imperfect page per hundred, on the hardest input this system has ever seen, under total distribution shift.

Part 1 ended with a rule: *count what escapes, and you're grading the system*. This was the rule's first live test, and it's the only reason I can write a sentence as precise as the one above instead of waving at an accuracy number.

The exam graded my grader too

One more thing the ambush caught — in my own tooling.

My first scoring pass penalized extracted rows for containing dose schedules like (1-0-1), even when they named the right drug. The scorer's precision check compared token sets in the wrong direction. External data debugs your evaluator, not just your model.

What I'm taking away

- **Reading skill doesn't travel; gate discipline does.** Every investment that held up under distribution shift was in the *system* — calibration, dual reads, the review queue. Everything invested in squeezing the reader stayed home.
- **My production target got context.** Part 1's benchmark sits at 96% trusted-output accuracy; this hostile set is unmeasurable against the 98% target. The real number for my actual target forms will land somewhere between the friendly bench and this ambush — and only the held-out set will say where.
- **A high flag rate is not failure.** 97% flags on unreadable input is the system telling the truth. The failure mode would have been 28% flags and a queue full of *Capelin*.

What's next

The held-out evaluation: real target-form registers, hand-filled, photographed, never published, created after every model's training cutoff. That produces the only numbers I'll stand behind — and it's part 3 of this series.

And the checkbox problem from part 1 still needs its own detection stage, likely powered by the one model I found that refuses to invent.

► **The numbers, for engineers (click to expand)**

This is part 2 of a series. Part 1: [Confidence Is Theater: What I Learned Building Scribe](#).

Code and methodology: github.com/kod201/scribe · My own benchmark gold set: [sohakanu/scribe-flow-gold](#)

All extraction ran locally on one machine; no page left it.